Literature search: queries

Methodology of Research, assignment 2 · Eduard Kecskemeti · topic: estimating “apparent weeks out” from physique photos (Classic Physique) with pairwise ranking · OpenAlex 17–19 Oct 2026, Scopus and Web of Science 7 Nov 2026

1. Search set-up

Item	Setting
Research question	How has progress toward contest leanness, or body composition in general, been estimated from ordinary photographs?
Sub-questions	Q1 body composition / shape from 2D images · Q2 body-composition tracking during contest preparation · Q3 learning an order or progression from images without absolute labels
Term refinement	Google Scholar Labs: each sub-question asked in natural language; results used only to collect vocabulary and seed papers
Databases	**OpenAlex** (api.openalex.org), field `title_and_abstract.search`; **Scopus**, field `TITLE-ABS`; **Web of Science Core Collection**, fields `TI` or `AB`. Scopus and WoS through the university library proxy. Google Scholar has no type/language filters or export.
Filters	OpenAlex: `publication_year:2016-2025, language:en, type:article|review|conference-paper|book-chapter` Scopus: `PUBYEAR > 2015 AND PUBYEAR < 2026 AND LANGUAGE(english) AND (DOCTYPE(ar) OR DOCTYPE(re) OR DOCTYPE(cp) OR DOCTYPE(ch))` WoS: `PY=(2016-2025) AND LA=(English) AND DT=(Article OR Review OR "Proceedings Paper" OR "Book Chapter")`
Criteria	Articles (incl. reviews), conference papers, book chapters; English; 2016–2025, current year (2026) excluded

2. Final queries (used for the dataset)

The same search string is used in all three databases; only the field syntax differs (section 5).

Block	Q	Query	OpenAlex	Scopus	WoS
b1v3	Q1	`("body fat" OR "body composition" OR adiposity OR "body shape" OR physique OR "body mass index" OR anthropometry OR anthropometric) AND (photograph OR photo OR "2D image" OR "single image" OR "RGB image" OR smartphone OR "computer vision" OR "image-based" OR silhouette OR "digital anthropometry") AND ("deep learning" OR "machine learning" OR "neural network" OR convolutional OR "artificial intelligence" OR "computer vision" OR automated OR "2D image" OR "two-dimensional")`	804	534	345
b2	Q2	`(bodybuilder OR bodybuilding OR "physique athlete" OR "physique competitor" OR "physique competition" OR "contest preparation" OR "competition preparation") AND ("body composition" OR "body fat" OR "fat loss" OR "weight loss" OR "energy restriction" OR "lean mass" OR "fat mass")`	294	176	137
b3v4	Q3	`("ordinal regression" OR "learning to rank" OR "pairwise ranking" OR "siamese network" OR "siamese neural network" OR "temporal order" OR "image ordering") AND (longitudinal OR progression OR "age estimation" OR "temporal change" OR "disease severity") AND (image OR images OR photograph OR imaging)`	138	90	75
b4	Q1–Q2	`(bodybuilding OR bodybuilder OR "physique athlete" OR "physique competitor") AND ("computer vision" OR "deep learning" OR "machine learning" OR "artificial intelligence" OR "pose estimation")`	18	10	9
		Unique records per database	**1,251**	**810**	**566**

Block b1 (the original Q1 query, 436 hits, see section 3) is a subset of b1v3 and adds no records; it is kept as a label on the records it matches.

3. Scrapped queries and why they were changed

Version	Query	Hits	Problem → change
Run 1 (all blocks)	Queries b1, b2 and b3 (original), with types `article|review|book-chapter` only	881 union	Conference papers are a separate OpenAlex type and were excluded; most ML work for Q1/Q3 is published at conferences → added `conference-paper`.
b1	`("body fat" OR "body composition" OR adiposity OR "body shape" OR physique OR "body mass index" OR anthropometry OR anthropometric) AND (photograph OR photo OR "2D image" OR "single image" OR "RGB image" OR smartphone OR "computer vision" OR "image-based" OR silhouette OR "digital anthropometry") AND ("deep learning" OR "machine learning" OR "neural network" OR convolutional)`	436 (302 in run 1)	Missed seed papers that describe the method without ML words (Majmudar 2022, Ferreira 2025) → third concept widened (b1v3).
b1 without 3rd concept	`("body fat" OR "body composition" OR adiposity OR "body shape" OR physique OR "body mass index" OR anthropometry OR anthropometric) AND (photograph OR photo OR "2D image" OR "single image" OR "RGB image" OR smartphone OR "computer vision" OR "image-based" OR silhouette OR "digital anthropometry")`	3,991	Recall fine but far over the 1,500 limit and mostly clinical / apparel noise → rejected.
b3 (original)	`("ordinal regression" OR "learning to rank" OR "pairwise ranking" OR "relative attribute" OR "temporal order" OR "image ordering" OR siamese) AND (longitudinal OR progression OR "disease stage" OR "temporal change" OR "age estimation") AND (image OR images OR photograph OR "deep learning")`	292	Bare *siamese* matched Thai/"Siam" studies; *progression* without an image term matched oncology → rewritten.
b3v2*	b3 with the third concept replaced by a required neural-network term ("deep learning" / "neural network" family)	46	Too narrow: lost known Q3 papers → rejected.
b3v3*	`("ordinal regression" OR "learning to rank" OR "pairwise ranking" OR "siamese network" OR "siamese neural network" OR "temporal order" OR "image ordering") AND (longitudinal OR progression OR "age estimation" OR "temporal change" OR "disease severity")`	703	Specific network phrases fixed the Siam noise, but with no image term most hits were non-imaging → image concept added (b3v4).
Scopus / WoS with keywords	The four final queries in `TITLE-ABS-KEY` (Scopus) and TS= (WoS, title + abstract + keywords)	Scopus 974 / 219 / 216 / 13 (1,422 unique); WoS 453 / 199 / 105 / 9 (766 unique)	Merged with OpenAlex: about 1,970 works, over the 1,500 limit → restricted to title and abstract, the same field OpenAlex searches.

** The exact wording of b3v2 was not preserved; it is described here from the search log. b3v3 is b3v4 without its image concept, as recorded in the log. The OpenAlex rows are run on 17–19 Oct; the Scopus/WoS keyword run on 7 Nov 2026.*

4. From hits to datasets (PRISMA-style flow)

Step	Records	Notes
Identified	2,631	OpenAlex 1,251 · Scopus 810 · Web of Science 566 · snowballing 4
After de-duplication	1,338	Records grouped by DOI, then by normalised title; 488 works are in all three databases
Cleaning, OpenAlex-only works	− 63	Rules R1–R7: spam adverts, retraction notices, front matter, no DOI and no venue, repository copies and preprints, non-English titles, duplicate titles
Cleaning, Scopus / WoS	− 6	D1 year 2026 (WoS early access) 1 · D2 retracted 3 · D3 same work as a removed OpenAlex record 1 · D4 encyclopedia entry 1
Dataset 1 (broad)	**1,269**	931 articles, 20 reviews, 283 conference papers, 35 book chapters; under the 1,500 limit. Found by: OpenAlex + Scopus + WoS 488 · OpenAlex + Scopus 239 · OpenAlex only 406 · OpenAlex + WoS 31 · Scopus + WoS 21 · Scopus only 60 · WoS only 20 · snowballing 4

Step	Records	Notes
Dataset 2 (relevant)	**50**	Screened from dataset 1; 42 found by OpenAlex, 4 only by Scopus/WoS, 4 from snowballing. Themes: photo-based body composition 15, shape/weight from images 8, contest prep 11, ranking/ordering 13, AI in bodybuilding 3

5. Database syntax (block b2 as the example)

OpenAlex (`title_and_abstract.search`, filters as a separate parameter):

```
(bodybuilder OR bodybuilding OR "physique athlete" OR "physique competitor" OR "physique competition" OR "contest preparation" OR "competition preparation") AND ("body composition" OR "body fat" OR "fat loss" OR "weight loss" OR "energy restriction" OR "lean mass" OR "fat mass")
```

Scopus advanced search:

```
TITLE-ABS((bodybuilder OR bodybuilding OR "physique athlete" OR "physique competitor" OR "physique competition" OR "contest preparation" OR "competition preparation") AND ("body composition" OR "body fat" OR "fat loss" OR "weight loss" OR "energy restriction" OR "lean mass" OR "fat mass")) AND PUBYEAR > 2015 AND PUBYEAR < 2026 AND LANGUAGE(english) AND (DOCTYPE(ar) OR DOCTYPE(re) OR DOCTYPE(cp) OR DOCTYPE(ch))
```

Web of Science advanced search (each concept group searched in title or abstract):

```
(TI=(bodybuilder OR bodybuilding OR "physique athlete" OR "physique competitor" OR "physique competition" OR "contest preparation" OR "competition preparation") OR AB=(bodybuilder OR bodybuilding OR "physique athlete" OR "physique competitor" OR "physique competition" OR "contest preparation" OR "competition preparation")) AND (TI=("body composition" OR "body fat" OR "fat loss" OR "weight loss" OR "energy restriction" OR "lean mass" OR "fat mass") OR AB=("body composition" OR "body fat" OR "fat loss" OR "weight loss" OR "energy restriction" OR "lean mass" OR "fat mass")) AND PY=(2016-2025) AND LA=(English) AND DT=(Article OR Review OR "Proceedings Paper" OR "Book Chapter")
```

All twelve exact strings are in search/queries.json in the GitHub repository.